

Sheet 1: DC Meter Movement

Kyrillos Refaat

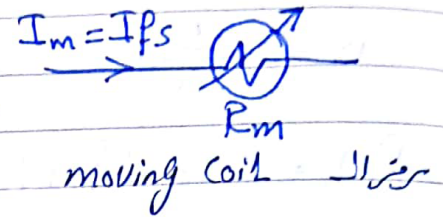
Chapter (1)

Direct-Current Indicating Instruments

1) DC Ammeter

R_m : Moving Coil Resistance

I_m : Full scale deflection current of the meter movement.

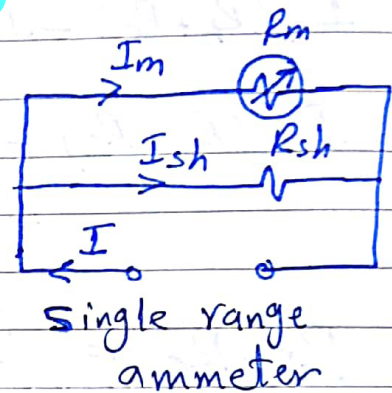


⇒ obtaining higher current scales:-

DC meter movement has a single current scale, but it can be used to measure larger currents if shunt resistor in parallel with the meter is used:

R_{sh} : Shunt resistance

I : Full-scale deflection current for the ammeter



$$V_m = I_m R_m$$

$$V_{sh} = V_m$$

$$I_{sh} = I - I_m$$

$$R_{sh} = \frac{V_{sh}}{I_{sh}} = \frac{V_m}{I_{sh}} = \frac{I_m R_m}{I_{sh}} = \frac{I_m R_m}{I - I_m} \quad \#$$

*** إذا ضربت n في التيار I_m يعطي التيار الكلي للتيار، الذي I (التي ليسب أضعاف).
 (التي ليسب أضعاف).

$$I = n I_m \quad \text{where } n: \text{multiplying factor}$$

$$R_{sh} = \frac{I_m R_m}{I - I_m} = \frac{I_m R_m}{n I_m - I_m} = \frac{R_m}{(n - 1)} \quad \#$$

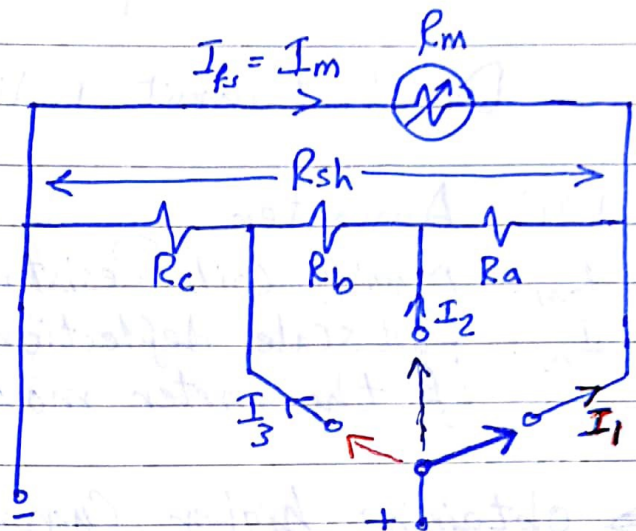
⇒ the Ayrton Shunt:

*** For I_1 -range:

$$R_{sh} = \frac{R_m}{(n-1)}$$

$$n = \frac{I}{I_m}$$

$$R_{sh} = R_a + R_b + R_c \quad (1)$$



*** For I_2 -range

$$V_{R_b+R_c} = V_{R_a+R_m} \Rightarrow (R_b + R_c)(I_2 - I_m) = I_m(R_a + R_m)$$

$$I_2(R_b + R_c) - I_m(R_b + R_c) = I_m(R_a + R_m)$$

$$I_2(R_b + R_c) = I_m(R_b + R_c) + I_m(R_a + R_m)$$

$$= I_m(R_a + R_b + R_c + R_m) = I_m(R_{sh} + R_m)$$

$$R_b + R_c = \frac{I_m(R_{sh} + R_m)}{I_2} \quad (2)$$

From (1), (2), we have $R_a = R_{sh} - (R_b + R_c) \quad \#$

*** For I_3 -range (maximum current range)

$$R_c = \frac{I_m(R_{sh} + R_m)}{I_3} \quad (3)$$

From (2) and (3), we have:

$$R_b = (2) - (3) = (R_b + R_c) - R_c \quad \#$$

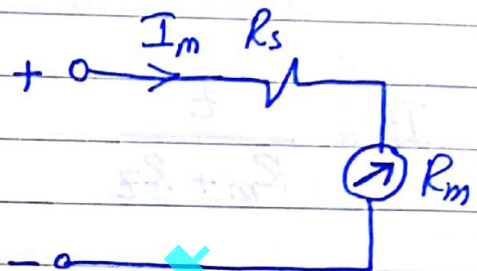
✗ ————— ✗

2) DC Voltmeter

the d'Arsonval meter movement can be converted to a DC voltmeter by connecting a multiplier resistor (R_s) series with the meter movement

R_s : multiplier resistor

$I_m = I_{fs}$ Full-scale current



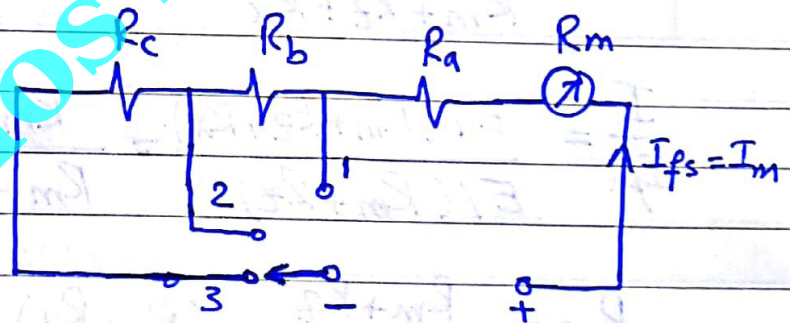
To Find the value of R_s :

① Determine the sensitivity (S): $S = \frac{1}{I_{fs}}$ Ω/V

② $R_s = S * \text{Range} - \text{Internal Resistance } (R_m)$

⇒ Multiple-range DC voltmeter::

$$S = \frac{1}{I_{fs}} \Omega/V$$



on Range (1): $R_a = S * \text{Range}_1 - R_m$

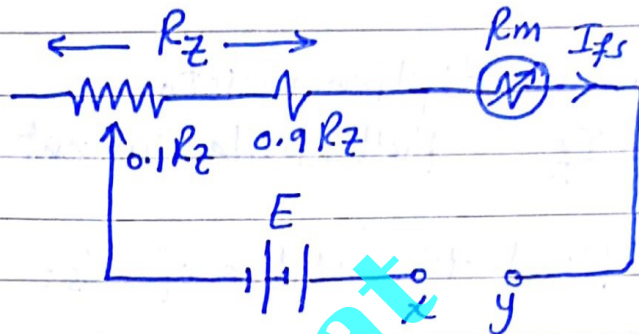
on Range (2): $R_b = S * \text{Range}_2 - (R_m + R_a)$

on Range (3): $R_c = S * \text{Range}_3 - (R_m + R_a + R_b)$

3) Ohmmeter:

the d'Arsonval meter movement may be used in conjunction with a battery and a resistor to construct a simple Ohmmeter circuit

$$I_{fs} = \frac{E}{R_m + R_z}$$



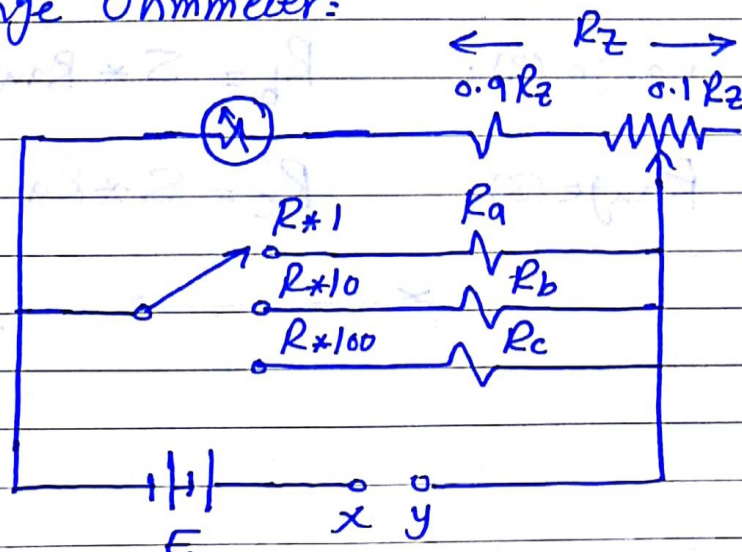
To determine the value of the unknown resistor (R_x) we connect it between points x and y.

$$I = \frac{E}{R_m + R_z + R_x}$$

$$\frac{I}{I_{fs}} = \frac{E / (R_m + R_z + R_x)}{E / (R_m + R_z)} = \frac{R_m + R_z}{R_m + R_z + R_x} = \rho$$

$$R_x = \frac{R_m + R_z}{\rho} - (R_m + R_z)$$

⇒ Multiple-range Ohmmeter:



1- A moving coil instrument has a resistance of $3k\Omega$ and gives full scale reading of 25 mA .calculate:

- The shunt resistance for a full scale deflection corresponding to 125A
- The series resistance for full scale reading with 625V. Find the power consumption in each case.

Solution:

(1) $R_m = 3k\Omega$, $I_m = 25\text{ mA}$

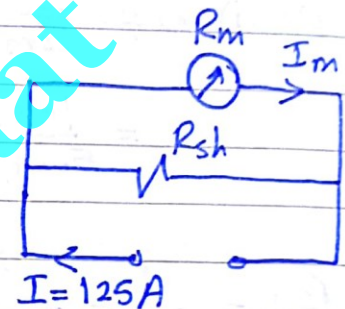
(a) $R_{sh} = ?$ $I = 125\text{ A (range)}$

(b) $R_s = ?$ $V = 625\text{ V (range)}$

Find the Power Consumption in each case.

(a) $n = \frac{I}{I_m} = \frac{125\text{ A}}{25 \times 10^{-3}\text{ A}} = 5000$

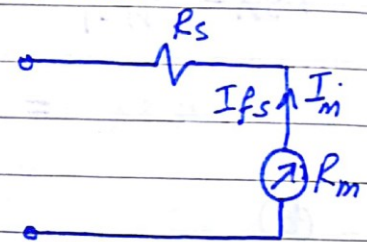
$R_{sh} = \frac{R_m}{n-1} = \frac{3k\Omega}{5000-1} = 0.6\Omega$



(b) $S = \frac{1}{I_{fs}} = \frac{1}{25 \times 10^{-3}} = 40\Omega/V$

$R_s = S \times \text{Range} - R_m$

$= 40 \times 625 - 3 \times 10^3 = 22000\Omega = 22k\Omega$



⇒ Power in each case:

$$P_{sh} = I_{sh}^2 \times R_{sh} = (I - I_m)^2 \times R_{sh}$$
$$= (125 - 25 \times 10^{-3})^2 \times 0.6\Omega$$
$$= 9371.25\text{ Watt}$$

$$P_s = I_m^2 \times R_s = (25 \times 10^{-3})^2 \times 22000$$
$$= 13.75\text{ Watt}$$

- 3- A moving coil am meter has the coil circuit resistance of $1k\Omega$. The range of the am meter is $0-500\mu A$.
- Calculate the value of shunt resistance to give a full scale deflection with a current of, (i) $10mA$, and (ii) $75mA$.
 - Find the value of shunt resistance if 40% deflection is obtained with a current of $100mA$.

Solution:

3) $R_m = 1k\Omega$ the range of ammeter $0-500\mu A$

(a) $R_{sh} = ??$ $I = (i) 10mA$ (ii) $75mA$

(b) $R_{sh} = ??$ if 40% deflection is obtained with current of $100mA$

(a) (i) $n = \frac{I}{I_m} = \frac{10mA}{500\mu A} = \frac{10 \times 10^{-3}}{500 \times 10^{-6}} = 20$

(ii) $n = \frac{75 \times 10^{-3}}{500 \times 10^{-6}} = 150$

$R_{sh} = \frac{R_m}{n-1} = \frac{1000}{20-1} = 52.63\Omega$

$R_{sh} = \frac{1000}{150-1} = 6.7\Omega$

(b)

40% $\rightarrow 100mA$
 100% $\rightarrow I$

Linear

40%
 100%

$I = 100mA$ $I = ??$

$I = \frac{100 \times 100mA}{40} = 0.25A$

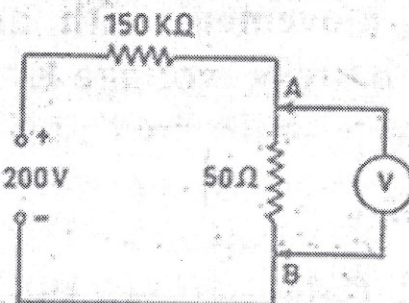
$n = \frac{I}{I_m} = \frac{0.25}{500 \times 10^{-6}} = 500$

$R_{sh} = \frac{R_m}{n-1} = \frac{1000}{500-1} = 2\Omega$

5- Two voltmeters of 50V range and having sensitivity of $1\text{k}\Omega/\text{V}$ and $20\text{k}\Omega/\text{V}$ respectively are used to measure voltage across $50\text{k}\Omega$ resistor in the circuit shown, calculate.

a) The reading of each voltmeter

b) The error in each reading expressed as a percentage of the true value.



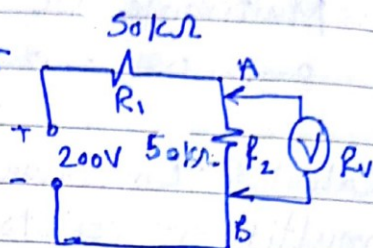
Solution:

(5) range = 50 V, $S_1 = 1\text{k}\Omega/\text{V}$, $S_2 = 20\text{k}\Omega/\text{V}$

(a) Calculate reading of each voltmeter

⇒ For S_1 with V_1

$$R_{V_1} = S_1 \times \text{Range} = 1 \frac{\text{k}\Omega}{\text{V}} \times 50\text{V} = 50\text{k}\Omega$$



$$R_{T1} = R_{V1} \parallel 50\text{k}\Omega = \frac{50\text{k}\Omega}{2} = 25\text{k}\Omega$$

$$V_1 = E \times \frac{R_{T1}}{R_{T1} + R_1} = 200 \times \frac{25}{25 + 150} = 28.57\text{V}$$

⇒ For S_2 with V_2

$$R_{V_2} = S_2 \times \text{Range} = 20 \frac{\text{k}\Omega}{\text{V}} \times 50\text{V} = 1\text{M}\Omega$$

$$R_{T2} = R_{V_2} \parallel 50\text{k}\Omega = \frac{1 \times 10^6 \times 50 \times 10^3}{10^6 + 50 \times 10^3} = 47.6\text{k}\Omega$$

$$V_2 = E \times \frac{R_{T2}}{R_{T2} + R_1} = 200 \times \frac{47.6}{47.6 + 150} = 48.19\text{V}$$

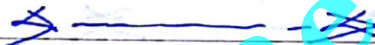
(b) Percentage error for each reading

$$V_{\text{exact}} = E * \frac{R_2}{R_1 + R_2} = 200 * \frac{50}{50 + 150} = 50 \text{ V}$$

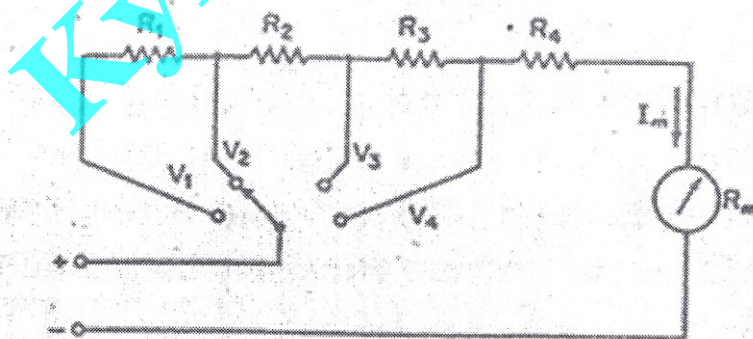
$$e_1 \% = 100 \% * \frac{\text{true} - V_1}{\text{true}} = \frac{50 - 28.57}{50} = 42.86 \%$$

$$e_2 \% = \frac{\text{true} - V_2}{\text{true}} * 100 \% = \frac{50 - 48.19}{50} * 100 \% = 3.62 \%$$

کما برادیت Sensitivity جهاز الفياسءء کما كانت القراءة
أقرب للقراءة المرحبةءء والخطأ من الفياسءء يقل



- 6- A basic d'Arsonval movement with internal resistance 100Ω and full scale current 1mA is to be converted into a multirange dc voltmeter with voltage ranges of $0-10\text{ V}$, $0-50\text{ V}$, $0-250\text{ V}$, and $0-500\text{ V}$. The circuit arrangement shown in figure is to be used for this voltmeter. Calculate the values of the multiplier resistors.

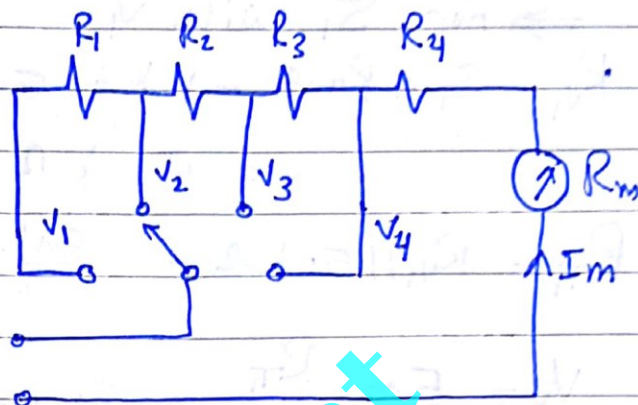


Solution:

(6) $R_m = 100\ \Omega$, $I_{fs} = 1\text{mA}$

Multirange Dc Voltmeter with voltage ranges
0-10, 0-50, 0-250V, 0-500V.

Calculate the values of
multiplier resistors.



$$S = \frac{1}{I_{fs}} = \frac{1}{1\text{mA}} = 1\text{k}\Omega/\text{V}$$

$$V_4 = 0-10\text{V} \quad V_3 = 0-50\text{V} \quad V_2 = 0-250\text{V} \quad V_1 = 0-500\text{V}$$

on the (0-10V = V_4) range

$$R_4 = S \times \text{Range}_4 - R_m = 1000 \times 10 - 100 = 9.9\text{k}\Omega$$

on the (0-50V) Range = V_3

$$R_3 = S \times \text{Range}_3 - (R_4 + R_m) = 1000 \times 50 - (9.9 \times 10^3 + 100) \\ = 40\text{k}\Omega$$

on the (0-250V) range = V_2

$$R_2 = S \times \text{Range}_2 - (R_3 + R_4 + R_m) \\ = 1000 \times 250 - (40 \times 10^3 + 9.9 \times 10^3 + 100) \\ = 200\text{k}\Omega$$

on the (0-500V) range = V_1

$$R_1 = S \times \text{Range}_1 - (R_2 + R_3 + R_4 + R_m) \\ = 1000 \times 500 - (200 + 40 + 9.9 + 0.1) \times 10^3 \\ = 250\text{k}\Omega$$

10- Calculate the value of the resistors R_1 through R_5 in the circuit of fig. P10.

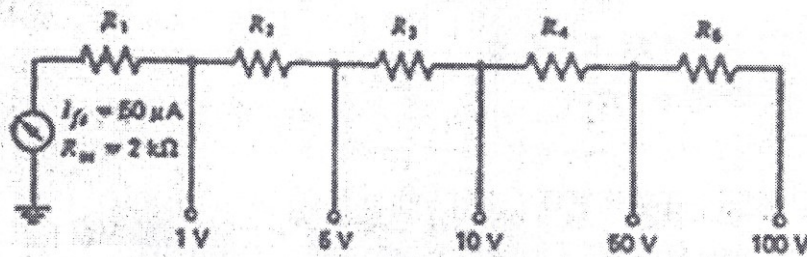


Fig. P10

Solution:

(10)

$R_m = 2\text{ k}\Omega$
 $I_{fs} = 50\text{ }\mu\text{A}$

$R_1 \rightarrow R_5 ??$

$$S = \frac{I_{fs}}{I_{fs}} = \frac{1}{50\text{ }\mu\text{A}} = 20\text{ k}\Omega/\text{V}$$

- on 1V Range:

$$R_1 = S \times \text{Range}_1 - R_m = 20 \frac{\text{k}\Omega}{\text{V}} \times 1\text{V} - 2\text{ k}\Omega = 18\text{ k}\Omega$$

- on 5V Range

$$R_2 = S \times \text{Range}_2 - (R_m + R_1) = 20 \times 5 - (2 + 18) = 80\text{ k}\Omega$$

- on the 10V Range:

$$R_3 = S \times \text{Range}_3 - (R_m + R_1 + R_2)$$

$$= 20 \times 10 - (2 + 18 + 80) = 100\text{ k}\Omega$$

- on the 50V Range:

$$R_4 = S \times \text{Range}_4 - (R_m + R_1 + R_2 + R_3)$$

$$= 20 \times 50 - (2 + 18 + 80 + 100) = 800\text{ k}\Omega$$

- on the 100V Range:

$$R_5 = S \times \text{Range}_5 - (R_m + R_1 + R_2 + R_3 + R_4)$$

$$= 20 \times 100 - (2 + 18 + 80 + 100 + 800)$$

$$= 1000\text{ k}\Omega = 1\text{ M}\Omega$$

- 8- Draw the schematic, including values, for an Ayrton shunt for a meter movement having a full scale deflection of 1mA and internal resistance of 500Ω to cover the current ranges of 1A , 5A and 10A .

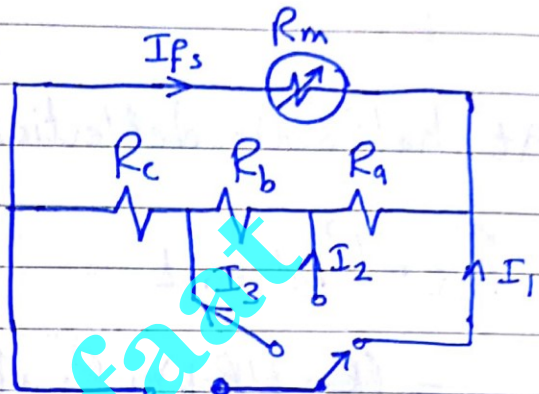
Solution:

(8) $I_{fs} = 1\text{mA}$, $R_m = 500\Omega$, Current ranges 1A , 5A , and 10A .

$$I_1 = 1\text{A}$$

$$I_2 = 5\text{A}$$

$$I_3 = 10\text{A}$$



For I_1 -range

$$n = \frac{I_1}{I_m} = \frac{1}{1 \times 10^{-3}} = 1000$$

$$R_{sh} = \frac{R_m}{n-1} = \frac{500}{100-1} = 0.5\Omega = R_a + R_b + R_c$$

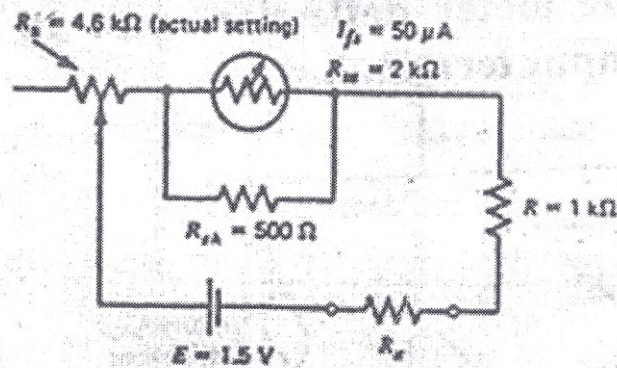
$$(R_b + R_c) = \frac{I_m}{I_2} (R_{sh} + R_m) = \frac{1 \times 10^{-3}}{5} (0.5 + 500) = 0.1\Omega$$

$$R_c = \frac{I_m}{I_3} (R_{sh} + R_m) = \frac{1 \times 10^{-3}}{10} (0.5 + 500) = 0.05\Omega \#$$

$$R_a = R_{sh} - (R_b + R_c) = 0.5 - 0.1 = 0.4\Omega \#$$

$$R_b = (R_b + R_c) - R_c = 0.1 - 0.05 = 0.05\Omega \#$$

- 9- In the circuit of fig. P9 what is the value of R_x if the meter reads half scale? Modify the circuit to multiple range ohmmeter and determine the values of R_x which give midscale deflect, with the assumed suitable values.



Solution:

(9) R_x ?? at half scale reading

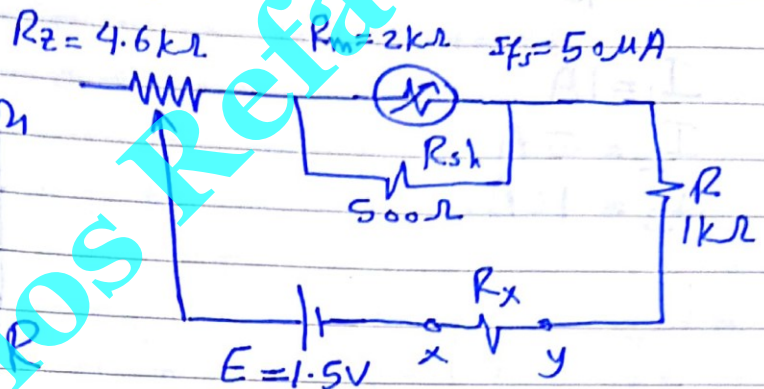
at half scale deflection

$$R_x = R_{\text{internal}}$$

$$= (R_m // R_{sh}) + R_z + R$$

$$= (2 // 0.5) + 4.6 + 1$$

$$R_x = 0.4 + 4.6 + 1 = 6 \text{ k}\Omega$$



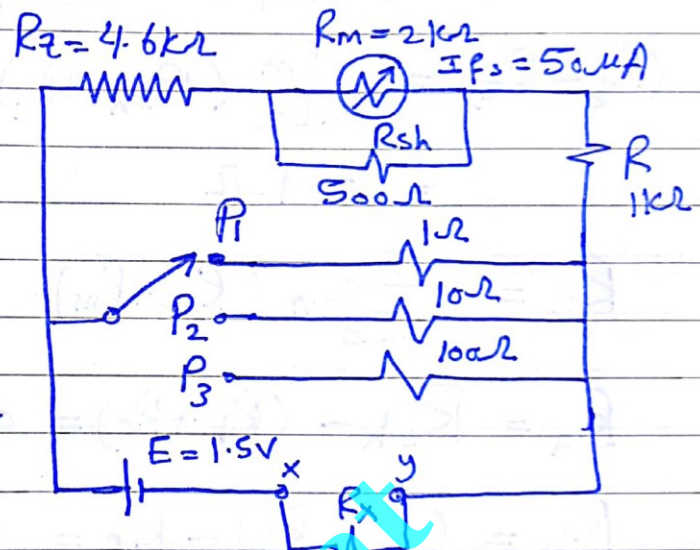
(b) Modify the circuit to multiple range ohmmeter
- at half scale deflection

$$R_x = R'_{\text{internal}}$$

⇒ at P_1 :

$$R'_{\text{int}} = 1 \parallel R_{\text{int}} = \frac{1 \times 6000}{6001} = 1 \Omega$$

$$R_x = 1 \Omega$$



⇒ at P_2 : $R'_{\text{int}} = 10 \parallel R_{\text{int}} = \frac{10 \times 6000}{6010} = 10 \Omega$
 $R_x = 10 \Omega$

⇒ at P_3 : $R'_{\text{int}} = 100 \parallel R_{\text{int}} = \frac{100 \times 6000}{6100} = 100 \Omega$
 $R_x = 100 \Omega$